ECE 3 Final Report

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Question 1: The Single Most Difficult Problem Description

The most difficult problem was finding a good pathfinding weave amount. Our team struggled a lot with understanding if the car was over-weaving, underweaving, and whether it was the effect of the Kd or Kp or our calibration.

Even when we isolated the Kp by itself, we didn't know what kind of weave we were looking for at a given speed to start adding Kd, nor what speed we should use as our "base" speed to start building up from. Another issue was choosing a good Kd/Kp ratio, since we knew that it was relatively car dependent and we also were unsure what the characteristics of a good ratio looked like on the track.

Overall, the difficulty of finding a good weave came from two parts. The first was finding a good "base" to start from, meaning what car speed, kd, kp, and kd/kp ratio we wanted to tune from. The second factor was our confusion in understanding how to identify if a change made progress or not. These two issues mean the majority of our track time was spent trying to tune kd kp.

Question 2: Solving the Most Difficult Problem

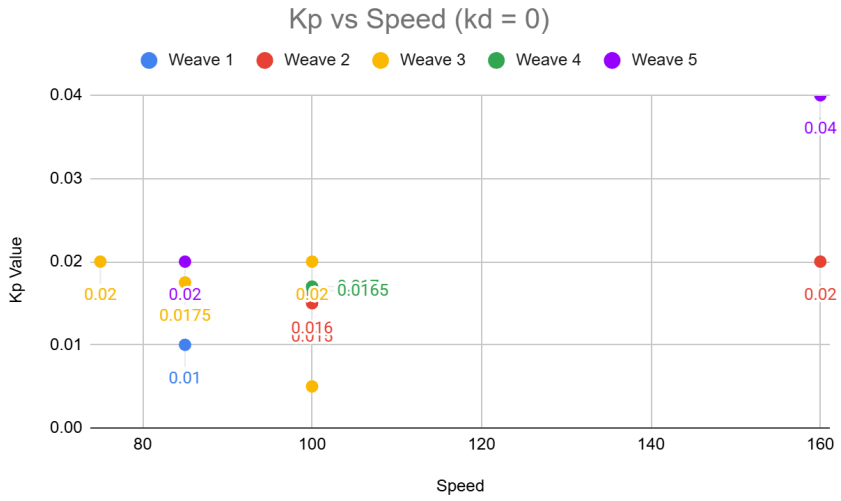
In order to solve the problem, we conducted a variety of tests in which we tried different weighting schemes for our sensors and altered the values of kp, kd, and our base speed. We started at a lower base speed of 75. At first we attempted to use the equation $k_p * maxerror = margin * maxspeeddifference$ to determine what our value of kp should be. Once we had a base value of 0.02, we built off of that to find kd, trying different ratios. Once the car was able to finish the track, we increased the base speed.

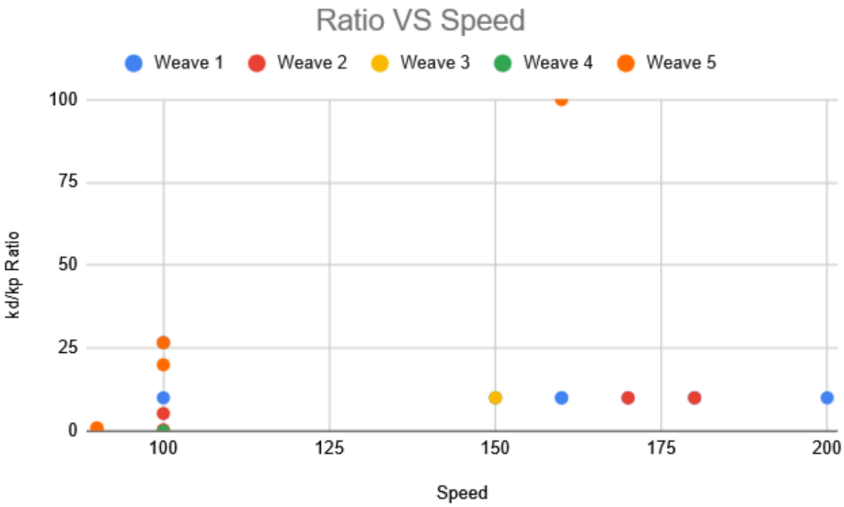
At higher base speeds, the car was much more erratic, and the values of kp and kd that had worked earlier were no longer correct. As a result, we decided to do more tests without kd to find a good kp value. We then added in kp, and realized that we needed to use a different ratio of kp/kd. After testing, we knew which kp values were too weak and which ones were too strong, but we couldn't find a good middle ground.

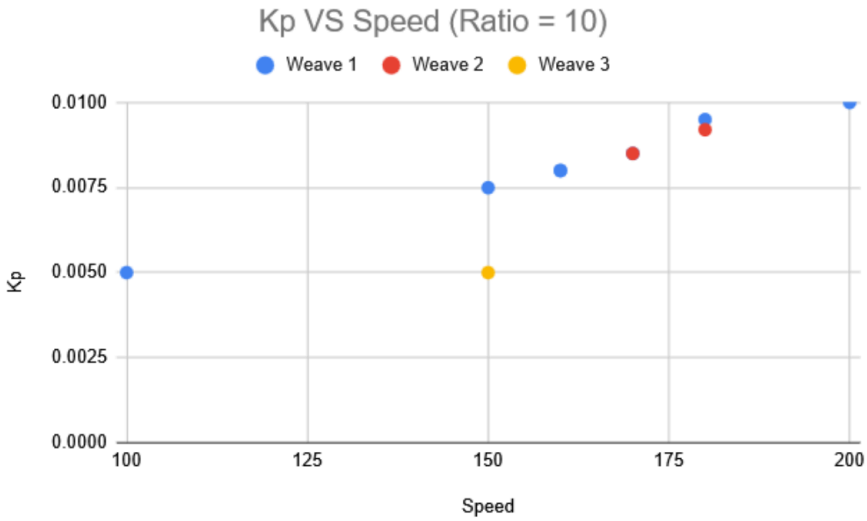
We realized that in order for the car to perform better at higher speeds, it needed to be more sensitive, so we did tests at different sensitivities to a slight curve and found that the car worked best at a high sensitivity.

Once the car was sensitive enough, we were able to adjust our kp and kd values until the car stopped weaving. After we found a ratio of kp to kd that worked, we were able to scale the same ratio to higher speeds.

Question 3: Analyzing the Most Difficult Problem Test Data







Question 4: Interpretation of the Graphs in Question 3

These charts show three conclusions which we came to while conducting our testing. The first chart shows how we came to find the optimal kp value for our car. We conducted a variety of different tests at the lower speed of 85 until we reduced the weaving of the car to 2-3 on a scale of 1 being the least weave and 5 being the most. At this amount of weave the car was able to finish the track. We then increased the speed to 100 and repeated the same process. We found that both times we converged on a value of kp between 0.01 - 0.02. Finally we increased the speed to our desired speed of around 160, before deciding to find a good kd value at a lower speed.

The second chart shows our attempt to find a good kd - kp ratio at different speeds. Our initial attempts to find a good kd at higher speeds did not work, so we tried out various different values at a speed of 100. Once we found one that worked, we realized that instead of finding good kp and kd values and generalizing that to higher speeds, we should keep the same ratio of kd/kp at higher speeds, while varying the actual value of kp and kd. We found that a good ratio was 10.

The final chart shows that when keeping the ratio of kd - kp the same, we could vary kp linearly in relation to the speed in order to ensure the weave was low. The values of kp we found were optimal were different from those we found to be optimal based on the earlier tests shown in chart 1 because of the changes in our car's sensitivity and the addition of kd. However, we were able to build off of the conclusion we came to in chart 2 to keep the ratio of kd - kp the same.

Question 5: Appendix

(1) ← no weaving

(5) ← too much veer off

ECE 3 Log Book

Kd	weave	Base Speed	Margin	Max Error	Start Position	End Position	Notes
	3	75	0.75	2800	1	12	
	3	75	0.75	2800	4	12	
	3	75	0.75	2800	3	12	15.675
0.01	2-3	75	0.75	2800	4	12	
0.1	3	75	0.75		4	12	
0.2	3	75	0.75		4	12	
1	3	75	0.75		4	12	
Kp*5	3	75	0.75		4	12	
Kp*10	5	75/100	0.75		4	0	turned in place
kp/10	5	100	0.75		4	4	
kp/10	5	100			4		
0	3	75	0.75		4	12	
0	5	85	0.75		4	3	
kp/5	3	85	0.75		4	12	
kp/3	5	85	0.75		4	3	
kp/3	4	85	0.65		4	12	
~~kp/2~~	~~5~~	~~85~~	~~0.65~~	kp	~~4~~		
			set kp directly				
kp/3	1	85		-0.01	4	1	did not correct
0	1	85		-0.01	4	1	↙ we stopped it
0	3	85		-0.0175	4	3	
kp/3		85		-0.0175	4		
kp/3	3	85		-0.02	4	12	
kp/2.5	3	85		-0.02	4	12	
kp/2.5	5	100		-0.02	4	3	
kp/2	5	100		-0.02	4	3	
0.01	5	100		-0.017	4	3	
0.01	2	100		-0.015	4	3	

Base speed	Kd	Kp	Weave	Start	End	Time	Notes	weights
100	0.01	0.02	2	4	3		did not turn	
100	0.004	0.02	2	4	3		↳	
100	0.004	0.02	2	4	3		↳	8 4 2 1/4
100	0.004	0.04	5	4	5		turned off track	↳
100	0.004	0.04	3	4	2		turned once but not enough	15 14 12 8/8
100	0.008	0.04	5	4	3		weaved off track	8 4 2 1/4
100	0.008	0.03	5	4	3		weaved off track	↳
100	0.008	0.02	2	4	3		did not turn	↳
100	0	0.02	2	4	3		↳	↳
100	0	0.02	2	4	3		↳	15 14 12 8/8
160	0	0.02	2	4	1		↳	↳
160	0	0.04	5	4	1		change battery went faster ↳	↳
160	0.008	0.04	?	4	1		↳	↳
160	4	0.04	5	4	3		turned off track	↳
90	0.4	0.04	5	4	4		↳	↳
90	0.04	0.04	5	4	5		↳	↳
90	0.02	0.03	5	4	4		↳ too wobbly sometimes turns	↳
100	0.4	0.02	5	4	1		change calibration a little	↳
100	0.4	0.02	5	1	1		original calibration	↳
100	0.4	0.015	5	1	1		overcorrects	↳
100	0.4	0.0015	1	1	4		too high kd	↳
100	0.04	0.0015	1	1	4		goes straight	↳
100	0	0.015	2	1	7		doesn't correct enough	↳
100	0	0.02	3	1			over corrects?	↳
100	0.09	0.017	2	1				↳
100	0	0.016	2	1			too less correcting	↳
100	0	0.017	4	1			too much	↳
100	0	0.0165	4	1			↳	↳

Base Speed	Kd	Kp	Weave	Start	End	Time	Notes	Weights
100	0	0.016	1	1	3		did not correct enough	17 14 12 8/8
								15 14 13 9/8
100	0	0.016	2	1	4			
100	0	0.02	3	1	4			16 14 10 2/2
100	0	0.005	5	1	4		better :D	16 14 10 4/2
100	0	0.005	3	1				
100	0.05	0.005	2	1	1		avoided track	
100	0.05	0.005	1	3	12		flip kD yayyy!!	
150	0.05	0.005	3	1				
150	0.075	0.0075	1	1	12		donut works	
150	0.075	0.0075	1	4	12		donut turns too much	
160	0.075	0.0075	1	4	24		takes too long to stop	
160	0.08	0.008	1	4	24		stops well	
200	0.1	0.01	1	4	2		goes off track	
160	0.08	0.008	1	1	24		15.8 secs	
160	0.08	0.008	1	3	24			
160	0.08	0.008	1	2	24			
160	0.08	0.008	1	4	24			
170	0.085	0.0085	1	3	12		increased donut speed did not do donut	
170	0.085	0.0085	1.5	3	24			
180	0.092	0.0092	2	3			veered off track	
180	0.095	0.0095	1	3	24		works :) 13.82	